

Type Ia Supernovae : Three-Dimensional Simulations of The Detonation Mechanism

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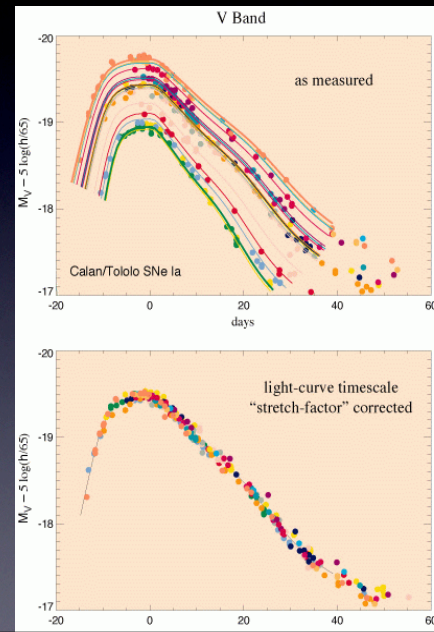
Outline

- I. Overview of Type Ia Supernovae
- II. First-Principles Simulation of Type Ia Supernovae

I. Overview of Type Ia Supernovae

Standardizing Light Curves - The Phillips Relation

- **Most** known Type Ia supernovae fit a universal light curve with a free parameter specified by the width of the curve. (Phillips, 1993)
- The Phillips relation allows astronomers to use Type Ia events as **standardizable candles** of cosmological distances, and the characterization of **dark energy** (Riess *et al*, 1998; Perlmutter *et al*, 1999)
- Residual scatter may be partially due to **prompt** and **delayed** Type Ia supernovae (Hamuy *et al*, 1996; Howell 2001; van den Bergh & Filippenko 2005)



White Dwarf Nuclear Energetics

- Fermi energy for N relativistic electrons in a fully degenerate white dwarf of radius R estimated by Pauli exclusion principle -

$$E_F \sim \hbar n^{1/3} c \sim \frac{\hbar N^{1/3} c}{R}$$

- Total energy is

$$E = N E_F + E_G \sim \frac{\hbar N^{4/3} c}{R} - \frac{G N^2 m_p^2}{R}$$

- Chandrasekhar mass is approximately

$$M_{\max} \sim \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{m_p^2} \sim \frac{m_{\text{Planck}}^3}{m_p^2} \sim 1.5 M_{\odot} \quad (1.4 M_{\odot})$$

- Energy release $^{12}\text{C} + ^{16}\text{O} \rightarrow ^{56}\text{Ni}$ is $8 \cdot 10^{17} \text{erg/g}$

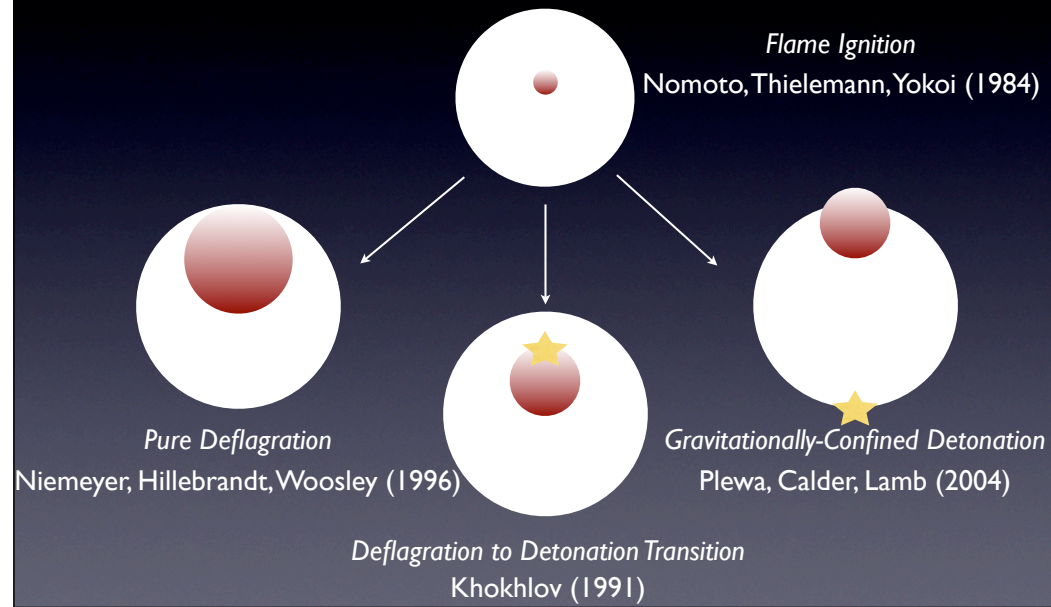
$$2.5 \cdot 10^{51} \text{erg} \sim 2.5 \text{ foe available}$$

Questions to be Addressed

- Is it possible to initiate a self-consistent detonation in Type Ia supernovae in three dimensions? What determines the success or failure of the mechanism?
- What are the observables associated with the mechanism? What are the isotopic abundances, and what is the nuclear yield?
- Are these models consistent with “typical” brightness Ia events?
- Can we begin to understand the nature of the variation in intrinsic luminosities in the prompt and delayed Type Ia channels?

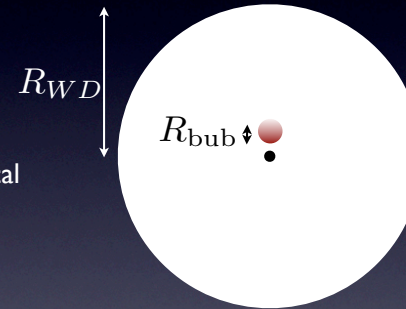
I. First-Principles Simulations of Ia Supernovae

Single Degenerate Type Ia Supernovae Mechanisms



First Successful 3-D Simulation of a Type Ia Detonation

- Background is a cold white dwarf model in initial equilibrium with initial mass 1.38 M_{sun} with a Helmholtz nuclear equation of state.
- Nuclear bubble is ignited within a spherical region slightly offset from the center of the white dwarf.
- Simulation numerically integrates the fully-coupled Euler, Poisson, and ADR PDEs forward in time.



Fundamental Equations

- Equations solved are Euler equations of hydrodynamics coupled to Poisson's equation for self-gravity and an advection-diffusion reaction model of combustion front :

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\mathbf{v} \rho) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\mathbf{v} \mathbf{v} \rho) = -\nabla P - \rho \nabla \Phi$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot [\mathbf{v} (\rho E + P)] = \rho \mathbf{v} \cdot \nabla \Phi + \rho \epsilon_{\text{nuc}}$$

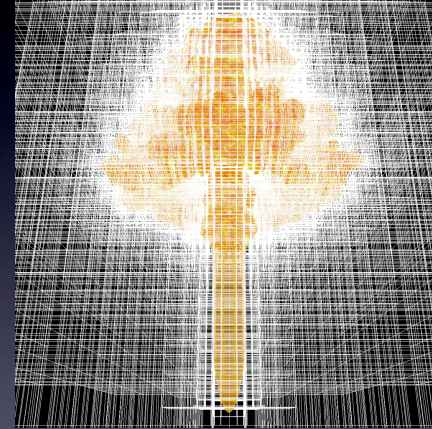
$$E = U + \frac{1}{2} \rho v^2$$

$$\nabla^2 \Phi = 4\pi G \rho$$

$$\frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi = \kappa \nabla^2 \phi + \frac{1}{\tau} R(\phi)$$

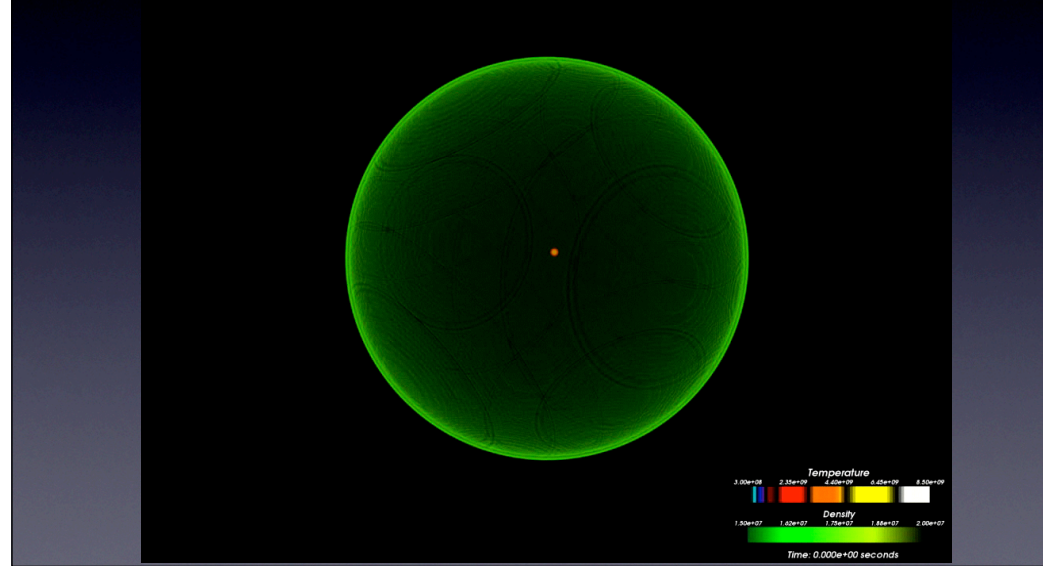
Adaptive Gridding

- Typical models have finest resolution equivalent to 700 cell zones across the initial diameter of the white dwarf progenitor, 8192 zones across the problem domain.
- With adaptive meshing, a single typical simulation followed from ignition to detonation requires roughly 5×10^4 CPU hours on current hardware; about 2 days of wall clock time on 10^3 CPUs.



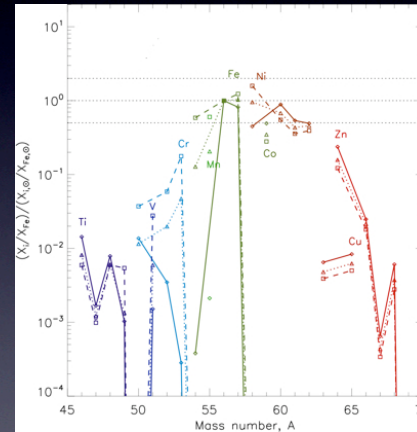
AMR Grid Refinement
of Flame Bubble

Animation of 3D Simulation of Single Bubble Type Ia Supernova GCD Through Detonation



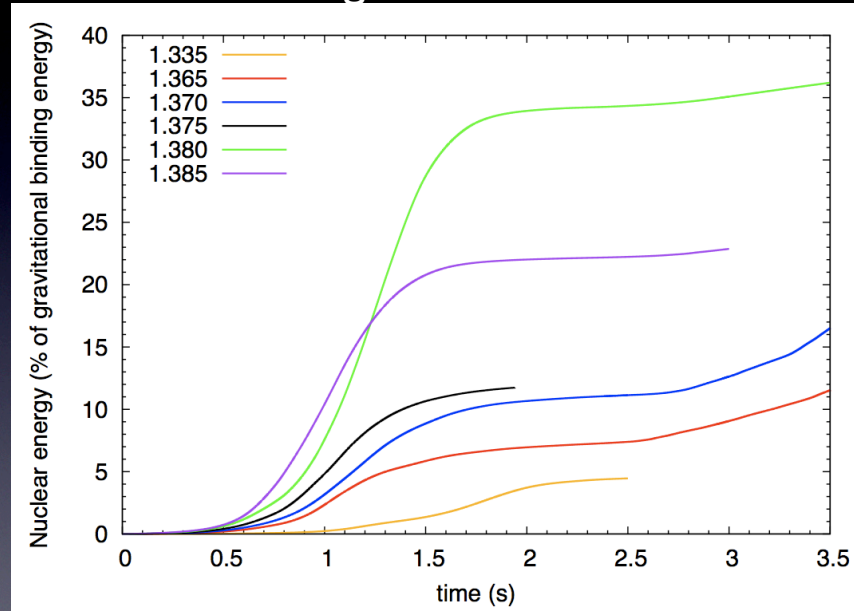
The Problem of Intermediate Mass Elements

- Initial models studied underproduced intermediate mass elements and overproduced Ni.
- Consequently, initial models were generally too luminous in comparison to Branch normal Ia events.
- We have subsequently explored the dependence of the nuclear energy release in the deflagration phase to the progenitor WD mass.



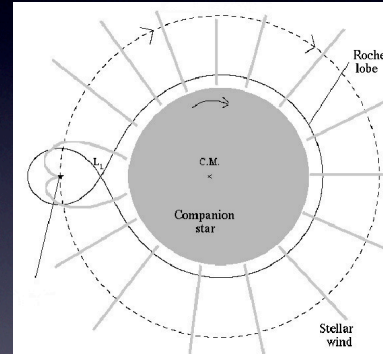
Meakin et al, 2009

Deflagration Energy Release Varying Progenitor Mass



A Model for Prompt and Delayed SNe Ia

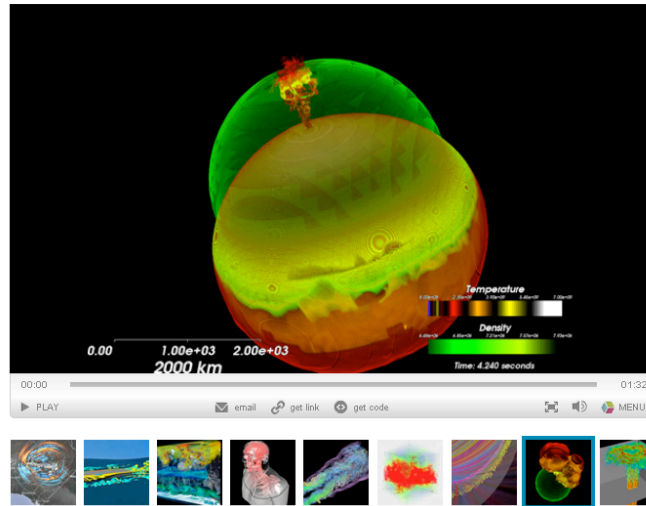
- Prompt SNe Ia with lifetimes $< 100 - 300$ Myr are observed to originate in star-forming galaxies; their companions are intermediate and high-mass stars with short lifetimes.
- Angular momentum wind-driven accretion transfer accretion from companion to WD progenitor is higher in the case of high-mass companions.
- The WD progenitors may therefore have a preferentially-higher spin and critical mass in prompt SNe Ia.
- These higher-mass WD progenitors are expected to have a greater Ni output, and therefore total luminosity, in the GCD model.



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Explosion of Type Ia Supernovae From Multiple Ignition Points

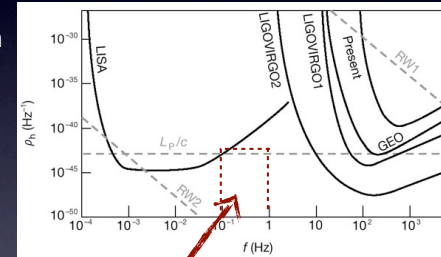
Type Ia supernovae are thought to be white dwarf stars in binary systems that explode due to a thermonuclear runaway. This movie shows a simulation of Type Ia supernovae exploding from multiple ignition points. When the hot ash breaks through the surface of the star, it spreads rapidly across the stellar surface, converges at the opposite point and produces a jet-like flow that triggers a detonation. The simulation shows that multiple ignition points generate more nuclear burning and produce more expansion of the star than a single ignition point. As a result, less radioactive nickel is produced during the detonation phase, and the explosion is less luminous.

Gravitational Wave Emission from SN Ia

- Type Ia events may emit a significant gravitational wave signal that may be detected by future space-borne missions.
- The characteristic frequency depends on the mean density at the instant of detonation :

$$f \propto \sqrt{G\rho}$$

- If detectable, GRW emission may establish a “smoking gun” for the GCD model.



Estimated SN Ia
at Virgo

Implications of 3D Simulations

- GCD model simulations demonstrate :
 - Successfull self-consistent detonations of Type Ia supernovae in 3D, without having to be initiated by hand.
 - Total Ni mass produced nearly independent of initial conditions of a single flame bubble.
 - Total Ni mass produced is sensitively dependent upon the progenitor mass and number of initial flame bubbles.
- Significant challenges remain :
 - A full understanding of the prompt and delayed channels for Type Ia supernovae may depend upon improved Ia progenitor models resulting from a full binary population synthesis.
 - Ignition of turbulent nuclear burning remains an open problem; models of the “simmering phase” prior to ignition are needed.